

BSIT 375: Knowledge Discovery and Data Science

Week 2 Lecture Notes: Multivariate Statistical Analysis
Based on Larose & Larose: *Discovering Knowledge in Data (2nd Ed)*

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1 Introduction to Multivariate Analysis

Why Multivariate Statistics?

In real-world data science problems, we rarely examine variables in isolation. Consider these scenarios:

- **Customer churn:** Affected by usage frequency, customer support interactions, subscription length, and demographics
- **Fraud detection:** Depends on transaction amount, location, time, user history, and multiple behavioral patterns
- **Product recommendations:** Based on purchase history, browsing behavior, ratings, and user similarities

Multivariate statistical analysis provides the toolkit to understand relationships between multiple variables simultaneously, enabling us to:

- Compare groups across multiple dimensions
- Quantify relationships between variables
- Make predictions while accounting for multiple factors
- Validate patterns discovered during exploratory data analysis

Key Concepts Before We Begin

Before diving into specific tests, let's establish fundamental concepts:

- **Population vs. Sample:** The population is the entire group we want to study; the sample is the subset we actually observe
- **Parameter vs. Statistic:** Parameters describe populations (μ , σ^2); statistics describe samples ($\bar{x}$, s^2)
- **Hypothesis Testing Framework:**
 - H_0 (Null hypothesis): No effect, no difference, status quo
 - H_a (Alternative hypothesis): There is an effect, a difference exists
 - α (Significance level): Probability of rejecting H_0 when it's actually true
 - Test Statistic: The Standardized value calculated from sample data to test H_0
 - p -value: Probability of observing our data if H_0 is true. If $p\text{-value} \leq \alpha$, we reject H_0 ; we fail to reject it.

2 5.1 Two-Sample t-Test for Difference in Means

When to Use This Test

The two-sample t-test answers questions like:

- "Do users who receive a new feature spend more time on the app than those who don't?"
- "Is the average transaction amount different between male and female customers?"
- "Do patients receiving a new drug have different recovery times compared to those on placebo?"

Theoretical Framework

Assumptions:

1. The two samples are independent
2. The data in each population are approximately normally distributed

Hypotheses: Let μ_1 and μ_2 be the population means for groups 1 and 2.

- Two-tailed test: $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 \neq \mu_2$
- One-tailed tests: $H_0 : \mu_1 \leq \mu_2$ vs. $H_a : \mu_1 > \mu_2$ (or reverse)

Test Statistics: Case 1: Equal Variances Assumed (Pooled t-test) When we believe $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we pool the variances:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \quad (1)$$

The test statistic is:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (2)$$

This follows a t-distribution with $df = n_1 + n_2 - 2$.

Case 2: Unequal Variances (Welch's t-test) In most data science applications, we cannot assume equal variances. Welch's t-test is more robust:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (3)$$

The degrees of freedom (ν) are approximated by the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}} \quad (4)$$

Examples

Scenario: A/B Testing for App Redesign

A mobile gaming company wants to test whether a new interface design increases daily active usage. They randomly assign users to:

- Group A (Control - Old Design): $n_1 = 40$ users
- Group B (Treatment - New Design): $n_2 = 35$ users

Data Collected (minutes spent per day):

Group A: $\bar{x}_1 = 25.3$ minutes, $s_1 = 5.1$ minutes

Group B: $\bar{x}_2 = 28.7$ minutes, $s_2 = 6.3$ minutes

Step 1: State Hypotheses

$H_0 : \mu_1 = \mu_2$ (No difference between designs)

$H_a : \mu_1 \neq \mu_2$ (Designs differ)

Step 2: Check Assumptions Sample sizes are moderate (40, 35). By Central Limit Theorem, the sampling distribution of means is approximately normal. We'll use Welch's test since variances appear different (5.1^2 vs 6.3^2).

Step 3: Calculate Test Statistic

$$\begin{aligned} t &= \frac{25.3 - 28.7}{\sqrt{\frac{5.1^2}{40} + \frac{6.3^2}{35}}} \\ &= \frac{-3.4}{\sqrt{\frac{26.01}{40} + \frac{39.69}{35}}} \\ &= \frac{-3.4}{\sqrt{0.65025 + 1.134}} \\ &= \frac{-3.4}{\sqrt{1.78425}} \\ &= \frac{-3.4}{1.336} \approx -2.545 \end{aligned}$$

Step 4: Calculate Degrees of Freedom

$$\begin{aligned}\nu &\approx \frac{(0.65025 + 1.134)^2}{\frac{(0.65025)^2}{39} + \frac{(1.134)^2}{34}} \\&= \frac{(1.78425)^2}{\frac{0.4228}{39} + \frac{1.285}{34}} \\&= \frac{3.1835}{0.01084 + 0.03779} \\&= \frac{3.1835}{0.04863} \approx 65.5 \quad (\text{use } df = 65)\end{aligned}$$

Step 5: Find Critical Value and p-value From t-table with $df = 65$, two-tailed $\alpha = 0.05$: Critical value $t_{0.025,65} \approx 1.997$

Our $|t| = 2.545 > 1.997$, so we reject H_0 .

Using statistical software, p -value ≈ 0.013 .

Step 6: Interpret Results The p -value (0.013) is less than $\alpha = 0.05$, providing strong evidence that the new design significantly changes user engagement. The negative t -value indicates Group B (new design) has higher mean usage ($28.7 > 25.3$), suggesting the redesign is effective.

3 5.2 Two-Sample Z-Test for Difference in Proportions

When to Use This Test

Compare proportions between two groups:

- "Is the click-through rate different between mobile and desktop users?"
- "Does the conversion rate vary between email campaign A and B?"
- "Is the proportion of defective products different between two production lines?"

Theoretical Framework

Assumptions:

1. Independent random samples
2. Large sample sizes: $n_1\hat{p}_1 \geq 10$, $n_1(1 - \hat{p}_1) \geq 10$ (same for group 2)
3. Binary outcomes (success/failure)

Test Statistic: Under $H_0 : p_1 = p_2 = p$, we estimate the common proportion:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} \quad (5)$$

where x_1 and x_2 are the number of successes in each group.

The test statistic is:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (6)$$

Example

Scenario: Email Marketing Campaign

A company tests two subject lines for their promotional email:

- Subject A: "50% Off Everything - Today Only!"
- Subject B: "Your Exclusive Discount Inside"

Results:

Group A: $n_1 = 5000$, $x_1 = 325$ opens $\hat{p}_1 = 0.065$
 Group B: $n_2 = 4800$, $x_2 = 412$ opens $\hat{p}_2 = 0.0858$

Step 1: Hypotheses

$H_0 : p_1 = p_2$ (No difference in open rates)
 $H_a : p_1 \neq p_2$ (Different open rates)

Step 2: Check Assumptions

$$\begin{aligned} n_1 \hat{p}_1 &= 5000(0.065) = 325 \geq 10 \\ n_1(1 - \hat{p}_1) &= 5000(0.935) = 4675 \geq 10 \\ n_2 \hat{p}_2 &= 4800(0.0858) = 412 \geq 10 \\ n_2(1 - \hat{p}_2) &= 4800(0.9142) = 4388 \geq 10 \end{aligned}$$

All conditions satisfied.

Step 3: Calculate Pooled Proportion

$$\begin{aligned} \hat{p} &= \frac{325 + 412}{5000 + 4800} = \frac{737}{9800} \approx 0.0752 \\ \hat{q} &= 1 - \hat{p} = 0.9248 \end{aligned}$$

Step 4: Calculate Standard Error and z-statistic

$$\begin{aligned} SE &= \sqrt{\hat{p}\hat{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \\ &= \sqrt{0.0752 \times 0.9248 \times \left(\frac{1}{5000} + \frac{1}{4800} \right)} \\ &= \sqrt{0.06955 \times (0.0002 + 0.0002083)} \\ &= \sqrt{0.06955 \times 0.0004083} \\ &= \sqrt{2.839 \times 10^{-5}} \approx 0.00533 \end{aligned}$$

$$z = \frac{0.065 - 0.0858}{0.00533} = \frac{-0.0208}{0.00533} \approx -3.90$$

Step 5: Determine Significance For $\alpha = 0.05$, critical values are ± 1.96 . $|z| = 3.90 > 1.96$, so reject H_0 .

$$p\text{-value} = 2 \times P(Z > 3.90) \approx 2 \times 0.000048 = 0.000096$$

Step 6: Interpret The extremely small p-value provides strong evidence that subject line B (0.0858) has a significantly higher open rate than subject line A (0.065). The company should use subject line B for future campaigns.

4 5.3 χ^2 Tests: Independence and Homogeneity

Conceptual Distinction

While these tests share a mathematical formula, they differ in their research design:

- **Test of Independence:** Determines if two categorical variables are associated within a single population.
- **Test of Homogeneity:** Determines if the proportions of a single categorical variable are consistent across multiple independent populations.

Theoretical Framework

Data Structure: Data are organized in an $r \times c$ contingency table:

- r = number of rows (groups or categories of variable A)
- c = number of columns (categories or categories of variable B)

Hypotheses (General Form):

- **Homogeneity:** H_0 : The distribution of the response variable is the same for all populations.
- **Independence:** H_0 : Variable A and Variable B are independent of one another.

Expected Frequencies (E_{ij}): Assuming H_0 is true, the expected frequency for each cell is:

$$E_{ij} = \frac{(\text{Row } i \text{ Total}) \times (\text{Column } j \text{ Total})}{\text{Grand Total (N)}} \quad (7)$$

Test Statistic: The chi-square statistic measures the "distance" between observed (O) and expected (E) data:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad (8)$$

This statistic follows a chi-square distribution with $df = (r - 1)(c - 1)$.

Worked Example: Customer Satisfaction

Scenario: A company wants to determine if customer satisfaction (Satisfied, Neutral, Dissatisfied) is the same across three regions (North, South, West). This is a **Test of Homogeneity** because we are comparing three distinct populations.

Step 1: Observed Frequencies (O_{ij}):

Region	Satisfied	Neutral	Dissatisfied	Total (R_i)
North	120	60	20	200
South	90	50	30	170
West	110	40	25	175
Total (C_j)	320	150	75	545 (N)

Step 2: Expected Frequency Calculations: For each cell, we apply $E_{ij} = \frac{R_i \times C_j}{N}$:

Region	Satisfied	Neutral	Dissatisfied
North	$\frac{200 \times 320}{545} = 117.43$	$\frac{200 \times 150}{545} = 55.05$	$\frac{200 \times 75}{545} = 27.52$
South	$\frac{170 \times 320}{545} = 99.82$	$\frac{170 \times 150}{545} = 46.79$	$\frac{170 \times 75}{545} = 23.39$
West	$\frac{175 \times 320}{545} = 102.75$	$\frac{175 \times 150}{545} = 48.17$	$\frac{175 \times 75}{545} = 24.08$

Step 3: Test Statistic Calculation:

$$\begin{aligned}\chi^2 &= \sum \frac{(O - E)^2}{E} = \frac{(120 - 117.43)^2}{117.43} + \cdots + \frac{(25 - 24.08)^2}{24.08} \\ &= 0.056 + 0.445 + 2.055 + 0.966 + 0.220 + 1.868 + 0.512 + 1.386 + 0.035 \\ \chi^2 &= \mathbf{7.543}\end{aligned}$$

Step 4: Conclusion: With $df = (3 - 1)(3 - 1) = 4$ and $\alpha = 0.05$, the critical value $\chi^2_{0.05,4} = 9.488$. Since $7.543 < 9.488$, we **fail to reject** H_0 . There is no evidence that satisfaction proportions differ by region.

5 5.4 Chi-Square Test for Goodness of Fit

When to Use This Test

Tests whether observed frequencies match an expected theoretical distribution:

- "Do customer purchases follow the expected 60-30-10 product category distribution?"
- "Is the die fair? (Are all numbers equally likely?)"
- "Does the website traffic follow the predicted 40% mobile, 35% desktop, 25% tablet distribution?"

Theoretical Framework

Hypotheses:

H_0 : The observed frequencies follow the specified distribution

H_a : The observed frequencies do not follow the specified distribution

Test Statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (9)$$

where k is the number of categories, and $E_i = n \times p_i$ with p_i being the hypothesized proportion for category i .

The test statistic follows a chi-square distribution with $df = k - 1 - m$, where m is the number of parameters estimated from the data (usually 0 for a fully specified distribution).

Example

Scenario: Website Traffic Sources

A website predicts its traffic should come from:

- Direct: 50%
- Search: 30%
- Social Media: 15%
- Referral: 5%

They collect data on 1000 visitors:

Direct: 480

Search: 320

Social: 140

Referral: 60

Step 1: State Hypotheses

H_0 : Traffic follows the predicted distribution

H_a : Traffic does not follow the predicted distribution

Step 2: Calculate Expected Frequencies

$$E_{\text{Direct}} = 1000 \times 0.50 = 500$$

$$E_{\text{Search}} = 1000 \times 0.30 = 300$$

$$E_{\text{Social}} = 1000 \times 0.15 = 150$$

$$E_{\text{Referral}} = 1000 \times 0.05 = 50$$

Step 3: Verify Assumptions All $E_i \geq 5$, condition satisfied.

Step 4: Calculate Chi-Square Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\text{Direct: } \frac{(480 - 500)^2}{500} = \frac{(-20)^2}{500} = \frac{400}{500} = 0.80$$

$$\text{Search: } \frac{(320 - 300)^2}{300} = \frac{20^2}{300} = \frac{400}{300} = 1.333$$

$$\text{Social: } \frac{(140 - 150)^2}{150} = \frac{(-10)^2}{150} = \frac{100}{150} = 0.667$$

$$\text{Referral: } \frac{(60 - 50)^2}{50} = \frac{10^2}{50} = \frac{100}{50} = 2.00$$

$$\chi^2 = 0.80 + 1.333 + 0.667 + 2.00 = 4.80$$

Step 5: Determine Degrees of Freedom and Critical Value $df = k - 1 = 4 - 1 = 3$
(no parameters estimated from data)

From chi-square table, $\chi^2_{0.05,3} = 7.815$

Step 6: Make Decision $\chi^2 = 4.80 < 7.815$, so fail to reject H_0 .

Step 7: Interpret Results The observed traffic distribution does not significantly differ from the predicted distribution. The company's traffic model appears valid.

6 5.5 Analysis of Variance (ANOVA)

When to Use ANOVA

ANOVA compares means across three or more groups:

- "Do customer satisfaction scores differ across four product versions?"
- "Is there a difference in average transaction amounts among five customer segments?"
- "Do three different training methods lead to different average productivity levels?"

Theoretical Framework

The Logic of ANOVA: ANOVA compares two sources of variation:

1. **Between-group variation:** How much do group means differ from the overall mean?
2. **Within-group variation:** How much do individual observations vary within their groups?

If the between-group variation is large relative to within-group variation, we have evidence that group means differ.

Partitioning the Total Variation:

$$\begin{aligned} \text{SST} &= \text{SSB} + \text{SSW} \\ \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 &= \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2 \end{aligned}$$

where:

- y_{ij} = observation j in group i

- $\bar{y}_{i.}$ = mean of group i
- $\bar{y}_{..}$ = overall mean
- n_i = sample size of group i
- k = number of groups

Mean Squares and F-Statistic:

$$MSB = \frac{SSB}{k - 1}$$

$$MSW = \frac{SSW}{n - k}$$

$$F = \frac{MSB}{MSW}$$

Under H_0 (all means equal), F follows an F-distribution with $df_1 = k - 1$, $df_2 = n - k$.

Complete Numerical Example

Scenario: Comparing Three Marketing Campaigns

A company tests three different advertising campaigns and measures weekly sales (in 1000s) :

Campaign A	Campaign B	Campaign C
12	15	10
14	16	12
11	14	11
13	18	9
10	17	13

Step 1: State Hypotheses

$H_0 : \mu_A = \mu_B = \mu_C$ (All campaigns have same mean sales)

H_a : At least one campaign mean differs

Step 2: Calculate Group Means and Overall Mean

$$\bar{y}_A = \frac{12 + 14 + 11 + 13 + 10}{5} = \frac{60}{5} = 12$$

$$\bar{y}_B = \frac{15 + 16 + 14 + 18 + 17}{5} = \frac{80}{5} = 16$$

$$\bar{y}_C = \frac{10 + 12 + 11 + 9 + 13}{5} = \frac{55}{5} = 11$$

$$\bar{y}_{..} = \frac{60 + 80 + 55}{15} = \frac{195}{15} = 13$$

Step 3: Calculate Sum of Squares

Between-group (SSB):

$$\begin{aligned}SSB &= \sum n_i(\bar{y}_{i.} - \bar{y}_{..})^2 \\&= 5(12 - 13)^2 + 5(16 - 13)^2 + 5(11 - 13)^2 \\&= 5(1) + 5(9) + 5(4) \\&= 5 + 45 + 20 = 70\end{aligned}$$

Within-group (SSW):

$$\text{Campaign A: } \sum (y - 12)^2 = (0)^2 + (2)^2 + (-1)^2 + (1)^2 + (-2)^2 = 0 + 4 + 1 + 1 + 4 = 10$$

$$\text{Campaign B: } \sum (y - 16)^2 = (-1)^2 + (0)^2 + (-2)^2 + (2)^2 + (1)^2 = 1 + 0 + 4 + 4 + 1 = 10$$

$$\text{Campaign C: } \sum (y - 11)^2 = (-1)^2 + (1)^2 + (0)^2 + (-2)^2 + (2)^2 = 1 + 1 + 0 + 4 + 4 = 10$$

$$SSW = 10 + 10 + 10 = 30$$

Total (SST):

$$SST = SSB + SSW = 70 + 30 = 100$$

Check: $\sum (y - 13)^2$ should equal 100.

Step 4: Calculate Mean Squares and F-Statistic

$$\begin{aligned}MSB &= \frac{70}{k - 1} = \frac{70}{2} = 35 \\MSW &= \frac{30}{n - k} = \frac{30}{15 - 3} = \frac{30}{12} = 2.5 \\F &= \frac{35}{2.5} = 14\end{aligned}$$

Step 5: Determine Critical Value $df_1 = 2$, $df_2 = 12$, $\alpha = 0.05$ From F-table, $F_{0.05,2,12} = 3.89$

Step 6: Make Decision $F = 14 > 3.89$, so reject H_0 .

Step 7: Post-hoc Tests Rejecting H_0 tells us at least one mean differs, but not which ones. We need post-hoc tests like Tukey's HSD:

Tukey's HSD (Honestly Significant Difference):

$$HSD = q_{\alpha,k,df} \times \sqrt{\frac{MSW}{n}} \quad (10)$$

where q is the studentized range statistic.

For our data, $q_{0.05,3,12} = 3.77$:

$$HSD = 3.77 \times \sqrt{\frac{2.5}{5}} = 3.77 \times \sqrt{0.5} = 3.77 \times 0.707 \approx 2.67$$

Compare differences:

$$\begin{aligned} |\bar{y}_B - \bar{y}_A| &= |16 - 12| = 4 > 2.67 && \text{Significant} \\ |\bar{y}_B - \bar{y}_C| &= |16 - 11| = 5 > 2.67 && \text{Significant} \\ |\bar{y}_A - \bar{y}_C| &= |12 - 11| = 1 < 2.67 && \text{Not Significant} \end{aligned}$$

7 5.6 Regression Analysis: Finding the Best-Fit Line

The Simple Linear Regression Model

$$Y_i = c + mX_i + e_i \quad (11)$$

where c is the intercept, m the slope, and e_i the error.

Ordinary Least Squares (OLS)

Minimize the sum of squared errors:

$$S(c, m) = \sum_{i=1}^n (Y_i - c - mX_i)^2 \quad (12)$$

Step 1: Partial Derivatives:

$$\frac{\partial S}{\partial c} = -2 \sum (Y_i - c - mX_i) = 0 \quad (13)$$

$$\frac{\partial S}{\partial m} = -2 \sum X_i(Y_i - c - mX_i) = 0 \quad (14)$$

Step 2: Normal Equations: From (13):

$$nc + m \sum X_i = \sum Y_i \quad (15)$$

From (14):

$$c \sum X_i + m \sum X_i^2 = \sum X_i Y_i \quad (16)$$

Step 3: Solve for m and c : From (15), $c = \bar{Y} - m\bar{X}$. Substitute into (16):

$$\begin{aligned} (\bar{Y} - m\bar{X}) \sum X_i + m \sum X_i^2 &= \sum X_i Y_i \\ m(\sum X_i^2 - \bar{X} \sum X_i) &= \sum X_i Y_i - \bar{Y} \sum X_i \end{aligned}$$

Recognising $\bar{X} \sum X_i = n\bar{X}^2$ and $\sum X_i Y_i - n\bar{X}\bar{Y} = \sum (X_i - \bar{X})(Y_i - \bar{Y})$,

$$m = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{SS_{XY}}{SS_{XX}} \quad (17)$$

Then

$$c = \bar{Y} - m\bar{X} \quad (18)$$

Example: Study Hours vs Exam Score

X (hours)	Y (score)
2	65
3	70
4	80
5	85
6	90

$$\begin{aligned}n = 5, \quad \sum X = 20, \quad \sum Y = 390, \quad \sum XY = 1625, \quad \sum X^2 = 90 \\ \bar{X} = 4, \quad \bar{Y} = 78, \quad SS_{XX} = 10, \quad SS_{XY} = 65 \\ m = 6.5, \quad c = 78 - 6.5 \times 4 = 52\end{aligned}$$

Thus $\hat{Y} = 52 + 6.5X$.

Key Properties of OLS

1. $\sum e_i = 0$.
2. The line passes through $(\bar{X}, \bar{Y})$.
3. $\sum X_i e_i = 0$.

8 5.7 Hypothesis Testing in Regression

Testing the Slope

In simple linear regression, the most critical question is: Does the predictor (X) actually help us predict the outcome (Y)? We test the population slope (β_1):

- $H_0 : \beta_1 = 0$ (No linear relationship; the line is flat).
- $H_a : \beta_1 \neq 0$ (A significant linear relationship exists).

Standard Error of the Estimate and the Slope

To know if our slope is significant, we first measure the "noise" or typical error in our model.

1. Residual Standard Error (s_e): This measures the standard deviation of the residuals (the vertical distances between data points and the regression line).

$$s_e = \sqrt{\frac{\text{Sum of Squared Errors (SSE)}}{n - 2}}, \quad \text{where } \text{SSE} = \sum (Y_i - \hat{Y}_i)^2 \quad (19)$$

Meaning: On average, how much do the actual Y values differ from the predicted $\hat{Y}$ values?

2. Standard Error of the Slope ($SE(\hat{\beta}_1)$): This measures how much the estimated slope would vary if we took different samples.

$$SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{\text{Sum of Squares for X (SS}_{XX})}} \quad (20)$$

The Test Statistic

We use a t -test to see how many standard errors the calculated slope is away from zero.

$$t = \frac{\text{Observed Slope} - \text{Hypothesized Slope}}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \quad (21)$$

The result is compared against a t -distribution with $df = n - 2$.

Example: Study Hours vs. Test Scores

Data: $n = 5$, $\hat{\beta}_1 = 3.25$, $SS_{XX} = 40$.

1. **Calculate SSE:** Based on the residuals $(0, -1.5, 2, 0.5, -1)$, the Sum of Squared Errors is 7.5.
2. **Calculate s_e :** $\sqrt{7.5/(5 - 2)} = \sqrt{2.5} \approx 1.581$.
3. **Calculate $SE(\hat{\beta}_1)$:** $1.581/\sqrt{40} \approx 0.25$.
4. **Calculate t :** $3.25/0.25 = 13$.

Decision: With $df = 3$, t_{crit} at $\alpha = 0.05$ is 3.182. Since $13 > 3.182$, we reject H_0 . Study hours significantly predict test scores.

9 5.8 Measuring Model Quality

The Coefficient of Determination (R^2)

The R^2 value (or "R-squared") tells us what percentage of the variation in Y is explained by our model.

Key Components:

- **Total Sum of Squares (SST):** The total variation in the actual Y values.
- **Sum of Squared Errors (SSE):** The variation that our model *failed* to explain (the residuals).

$$R^2 = 1 - \frac{\text{Unexplained Variation (SSE)}}{\text{Total Variation (SST)}} \quad (22)$$

Example Calculation: Using our Study Hours data where $SST = 430$ and $SSE = 7.5$:

$$R^2 = 1 - \frac{7.5}{430} = 1 - 0.0174 = 0.9826 \quad (23)$$

Interpretation: 98.26% of the variation in test scores is explained by the number of hours studied.

The Correlation Coefficient (r)

The correlation coefficient measures the strength and direction of the linear relationship. It is the square root of R^2 , with the sign (+/−) matching the slope.

$$r = \sqrt{0.9826} \approx 0.991 \quad (24)$$

A value of 0.991 indicates a very strong, positive linear relationship.

10 5.12 Multiple Regression

Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i \quad (25)$$

Matrix Form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (26)$$

Example: House Prices

Predict price (Y , in \$100k) from size (X_1 , in 1000 sq ft) and bedrooms (X_2). Data:

Y	X_1	X_2
2.5	1.5	3
3.0	2.0	4
2.8	1.8	3
3.5	2.2	4
3.2	2.0	3

Solving gives $\hat{Y} = 0.5 + 0.8X_1 + 0.2X_2$.

Multicollinearity

High correlation among predictors inflates standard errors. Detected by Variance Inflation Factor (VIF):

$$VIF_j = \frac{1}{1 - R_j^2}, \quad R_j^2 \text{ from regressing } X_j \text{ on the other predictors} \quad (27)$$

$VIF > 10$ indicates serious multicollinearity.

Adjusted R^2

Penalizes adding unnecessary predictors:

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1} \quad (28)$$